

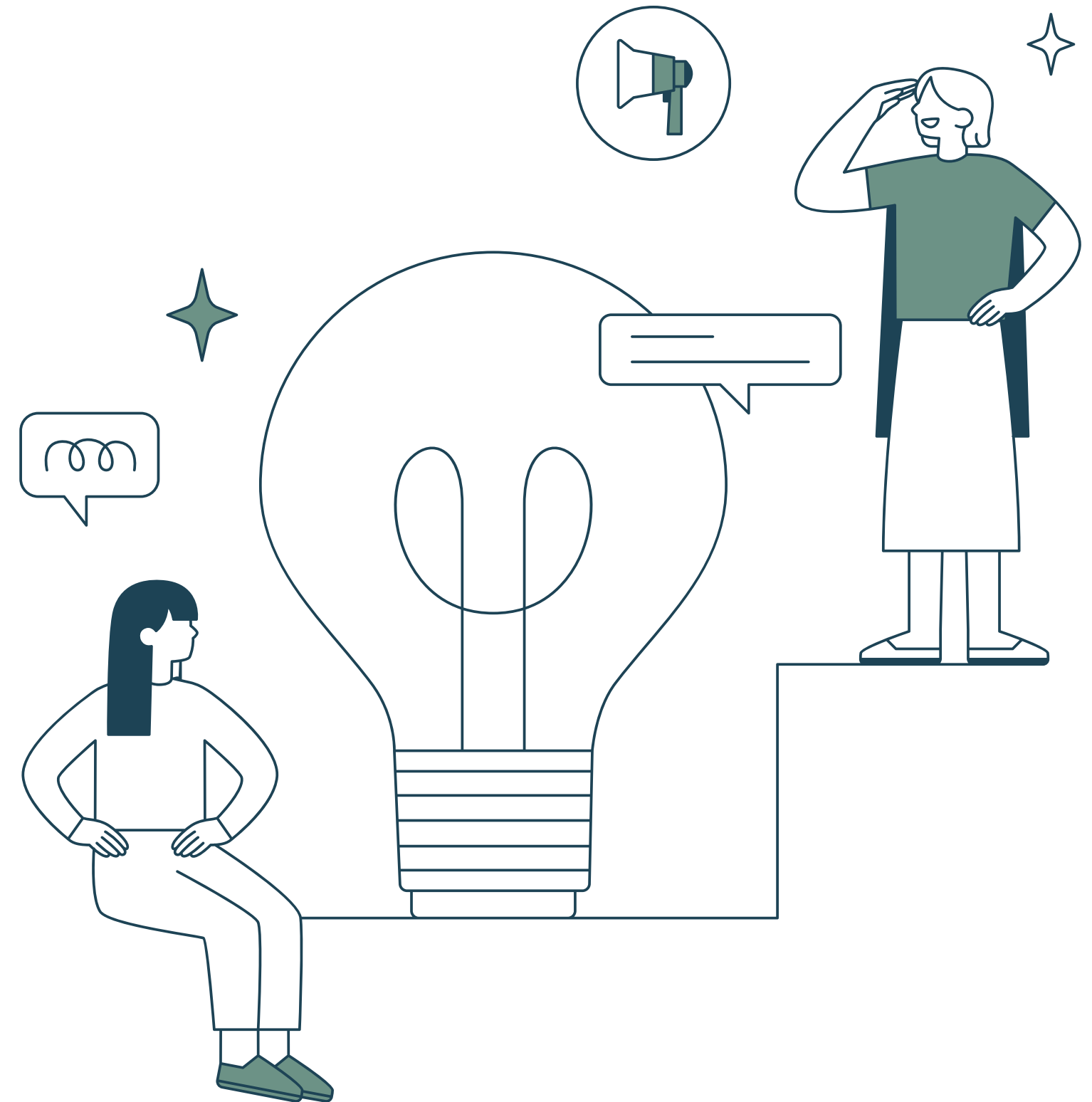
E-wallet Analysis Report

Droppping Bill Payment Success Rate



Outline

- 01** Executive Summary
- 02** Root Cause Analysis
- 03** Recommendations

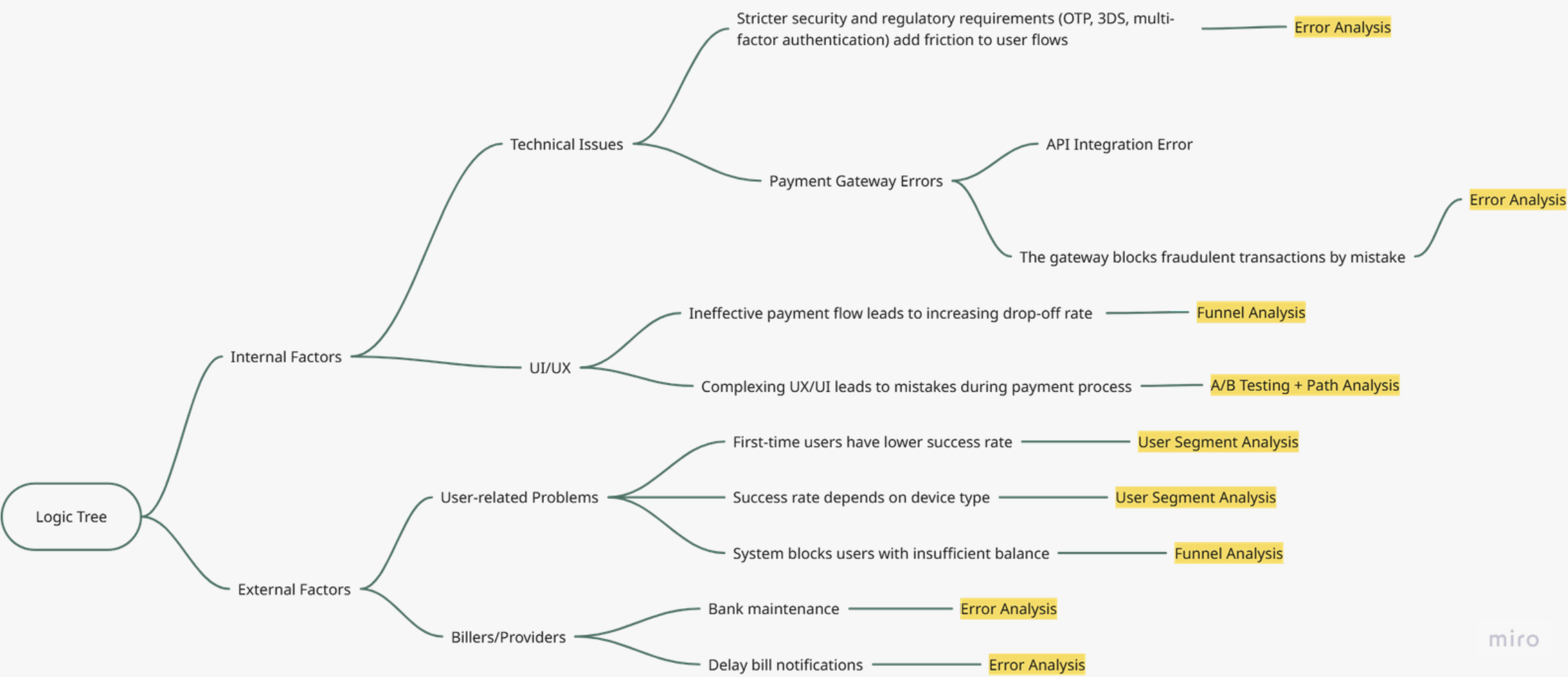


Executive Summary

SITUATION Business Context	• Transactions increased after the 3.9.6 update on July 28th . • Billing payment transaction accounts for the greatest share (37%).
COMPLICATION Business problems	• Declining Success Rates: Overall payment success rate dropped by 5% between July and August. • Authentication Bottlenecks: High drop-off rates driven by 3DS and OTP timeouts, specifically during the authentication phase. • Card Performance Issues: Credit and Debit cards underperformed other methods with a peak 14% error rate. • Segment Risk: Dormant users experienced disproportionately higher error rates compared to active users.
QUESTIONS Business concerns	• How can we stabilize the backend infrastructure and optimize the authentication flow to reduce the 18% drop-off? • How can we successfully re-engage our dormant and user segments aged 35-44? • How can we improve the customer support system with in-time communication and insightful information?
ANSWERS Business solutions	• Reduce authentication failures and backend-related drop-offs. • Payment flow optimization & Root cause validation by performing A/B testing and comparison between versions. • Improving real-time customer support and re-engaging with dormant users through marketing campaigns.

Hypothesis Breakdown

Using the Issue Tree, we hypothesized possible factors that led to the dropping success rate, including Internal and External factors.



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01.

Root Cause Analysis

- Analyze data to identify key problems.
- Draw insights on key factors to develop solutions.

02.

Recommendations

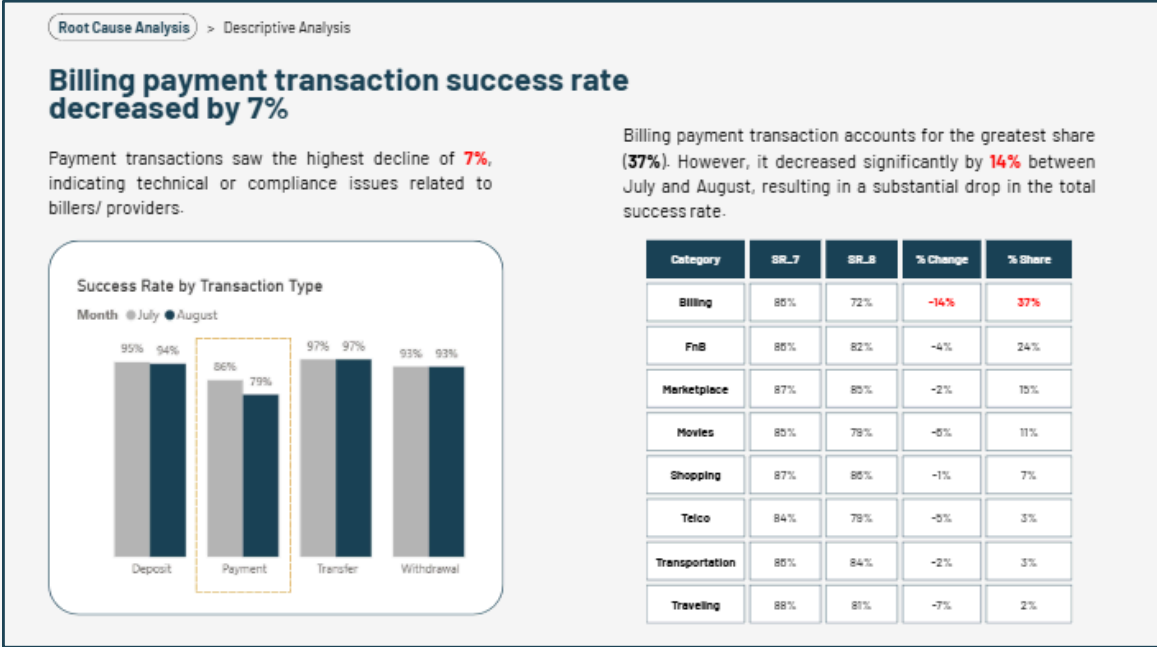
- Based on the insights, propose solutions that directly improve the success rate in the following months.
- Highlight important features that need to be tracked and tested.

Root Cause Analysis Overview

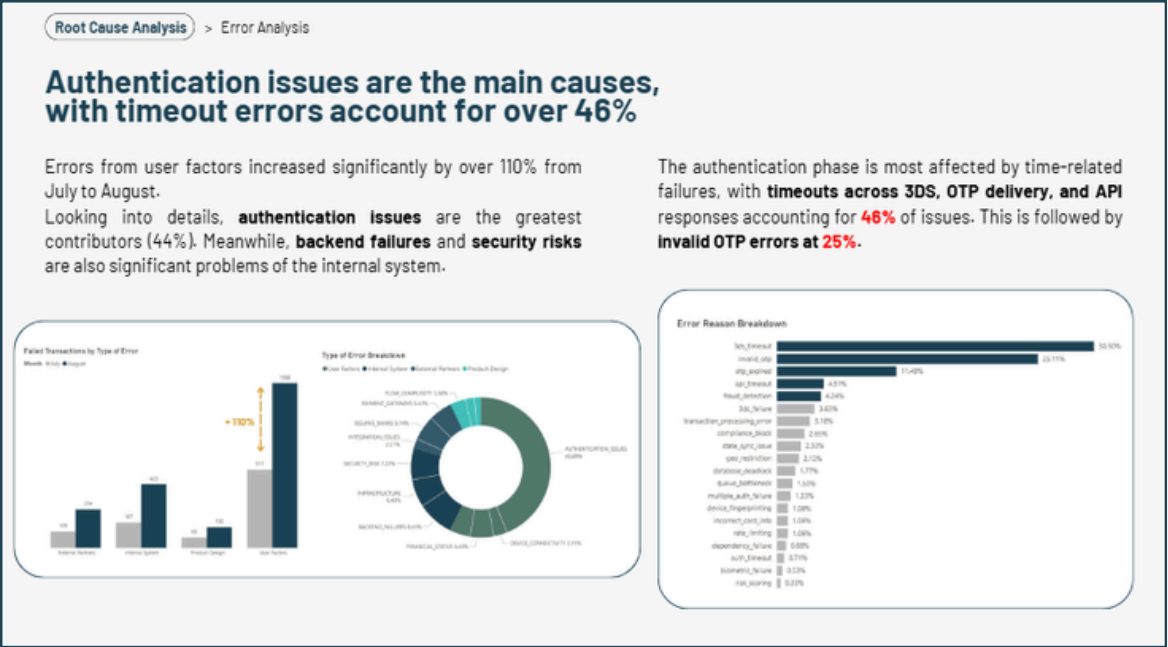
01.

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Errors from billing payment transactions contributed significantly to the increasing error rate in August, mainly because of version updates from July 28th onwards.



Breaking down the main errors, authentication issues accounted for the highest percentage. Specifically, there was an increase in 3D and OTP errors, including timeouts and invalid ones.

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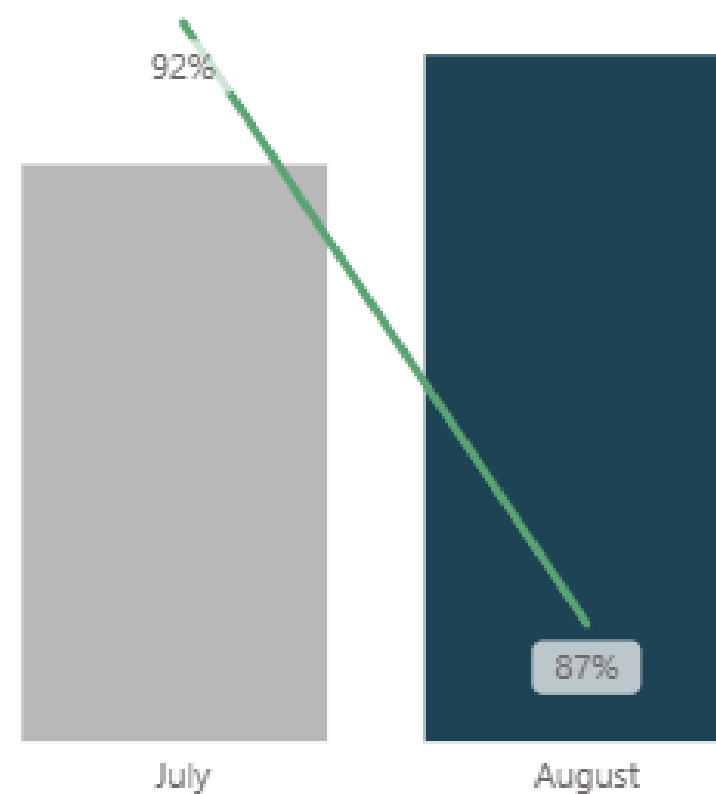
The payment success rate declined despite an increase in total transactions

The payment success rate **declined from 92% to 87%**, although the number of total transactions increased by approximately **20%** between July and August 2023.

There was a sharp increase in transactions on July 28th, followed by a more intense rise until August 8th. However, the **success rate** at the end of the period remained **lower** than in the first half of July.

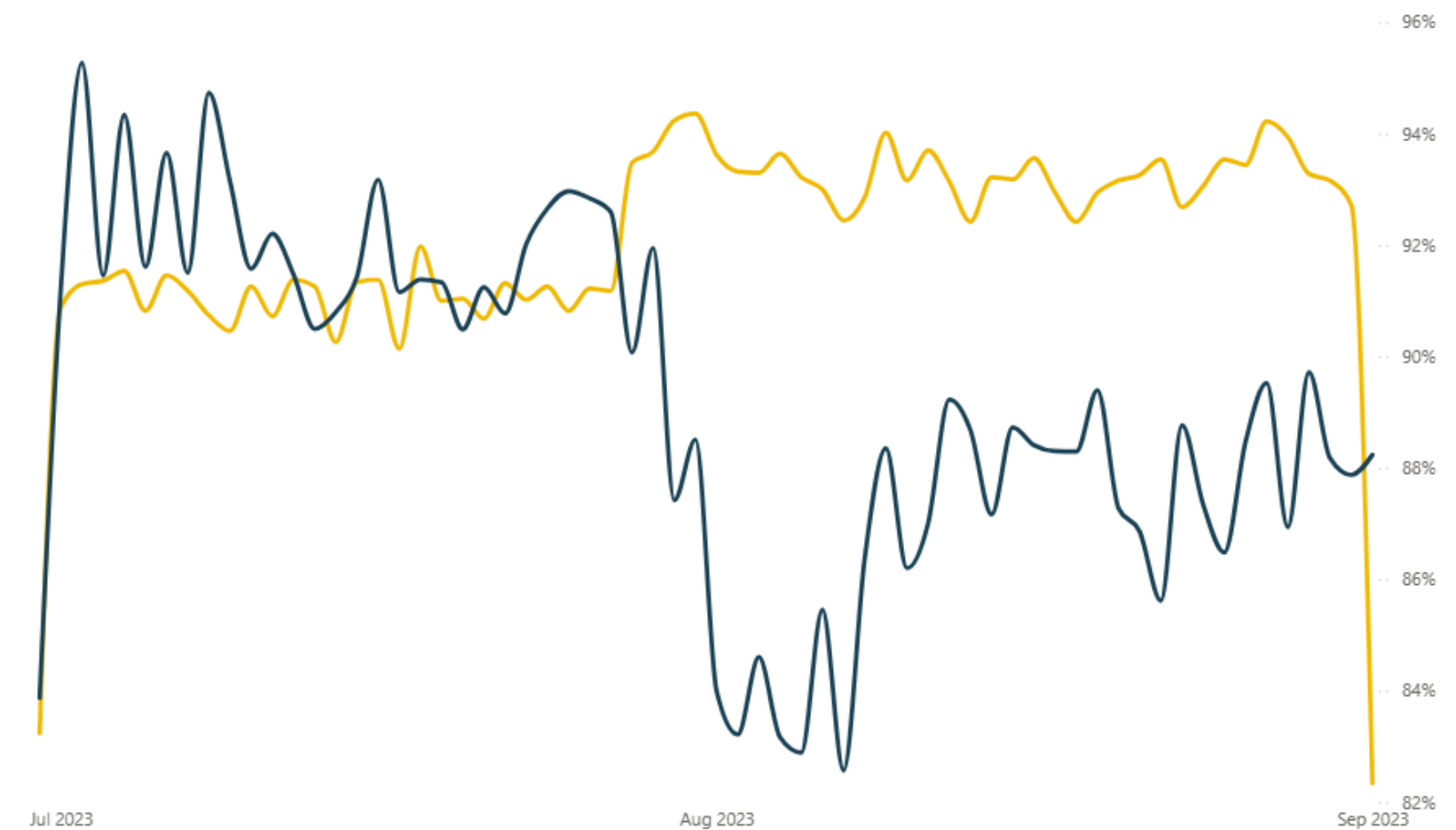
Success Rate by Month

● Number of transactions ● Success Rate



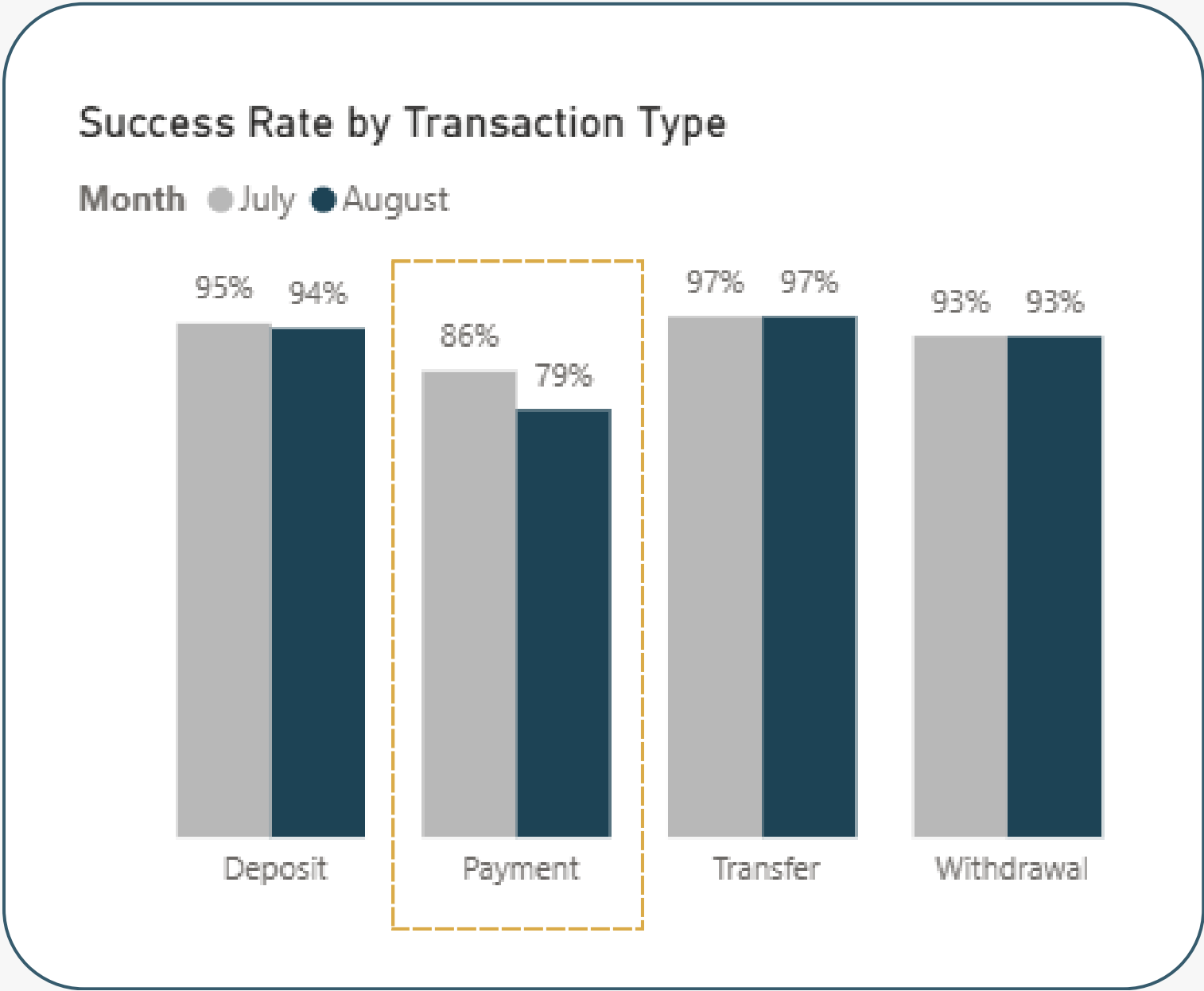
Transactions and Success Rate by Date

● Transactions ● Success Rate



Billing payment transaction success rate decreased by 7%

Payment transactions saw the highest decline of **7%**, indicating technical or compliance issues related to billers/ providers.



Billing payment transaction accounts for the greatest share (**37%**). However, it decreased significantly by **14%** between July and August, resulting in a substantial drop in the total success rate.

Category	SR_7	SR_8	% Change	% Share
Billing	86%	72%	-14%	37%
FnB	86%	82%	-4%	24%
Marketplace	87%	85%	-2%	15%
Movies	85%	79%	-6%	11%
Shopping	87%	86%	-1%	7%
Telco	84%	79%	-5%	3%
Transportation	86%	84%	-2%	3%
Traveling	88%	81%	-7%	2%

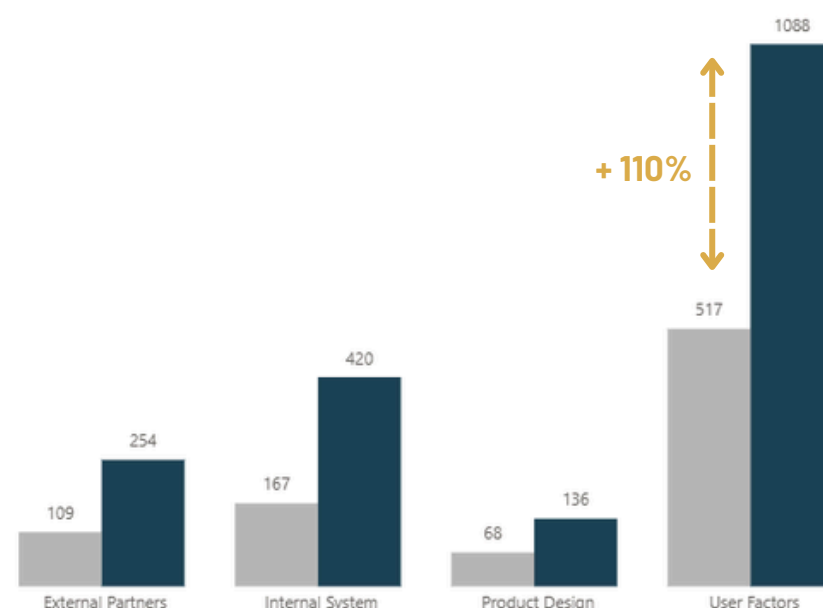
Authentication issues are the main causes, with timeout errors contributing over 46%

- Errors from user factors increased significantly by (+110%) from July to August.
- Authentication issues** are the greatest contributors (44%). Meanwhile, **backend failures** and **security risks** are also significant problems with the internal system.

The authentication phase is most affected by time-related failures, with **timeouts across 3DS, OTP delivery, and API responses** accounting for **46%** of issues. This is followed by **invalid OTP errors at 25%.**

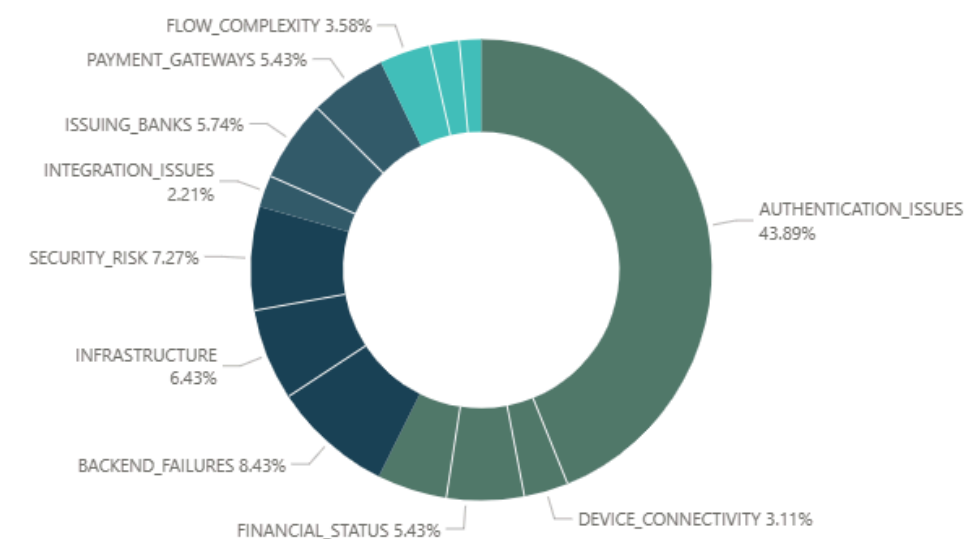
Failed Transactions by Type of Error

Month ● July ● August

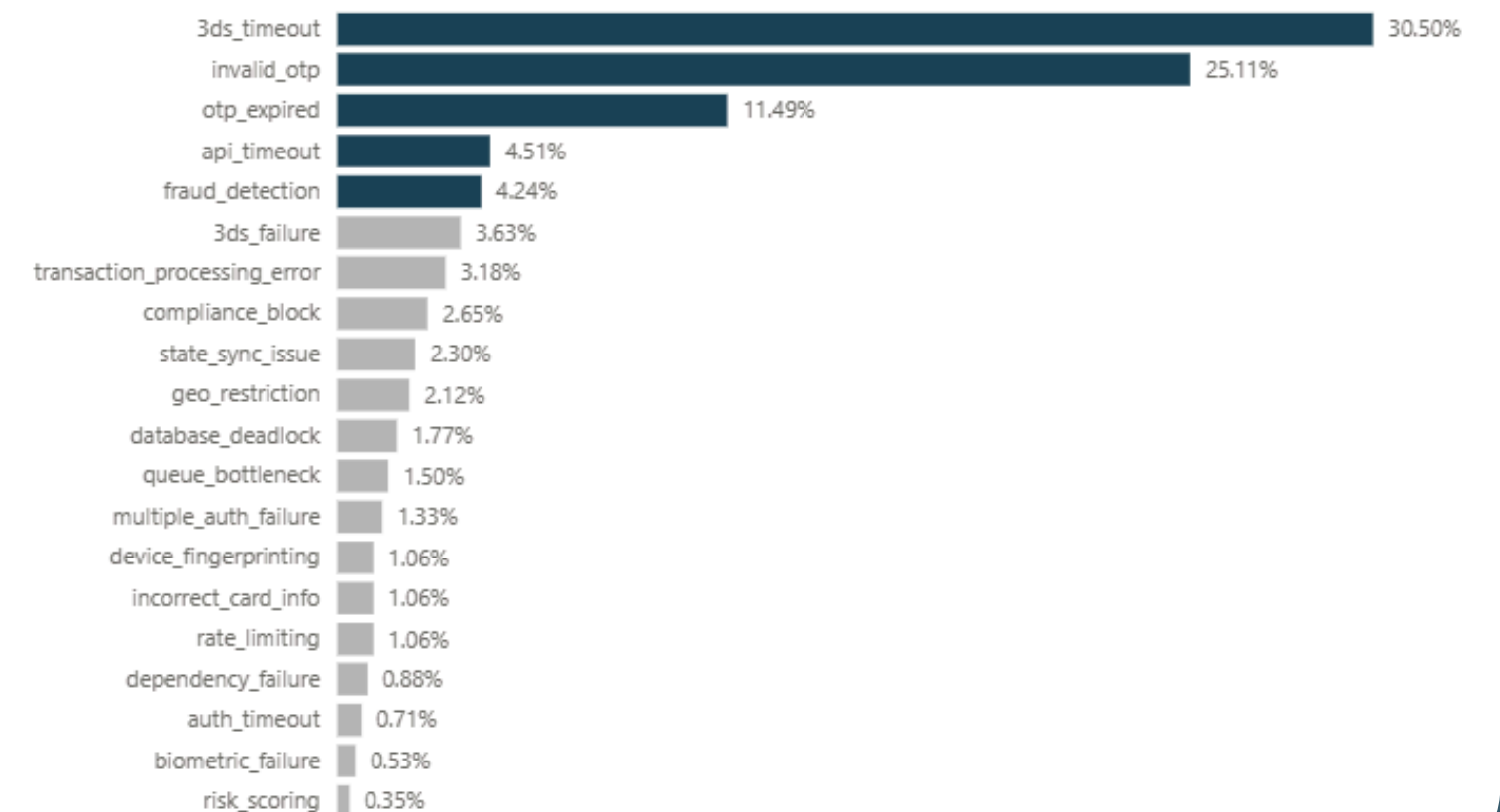


Type of Error Breakdown

● User Factors ● Internal System ● External Partners ● Product Design



Error Reason Breakdown

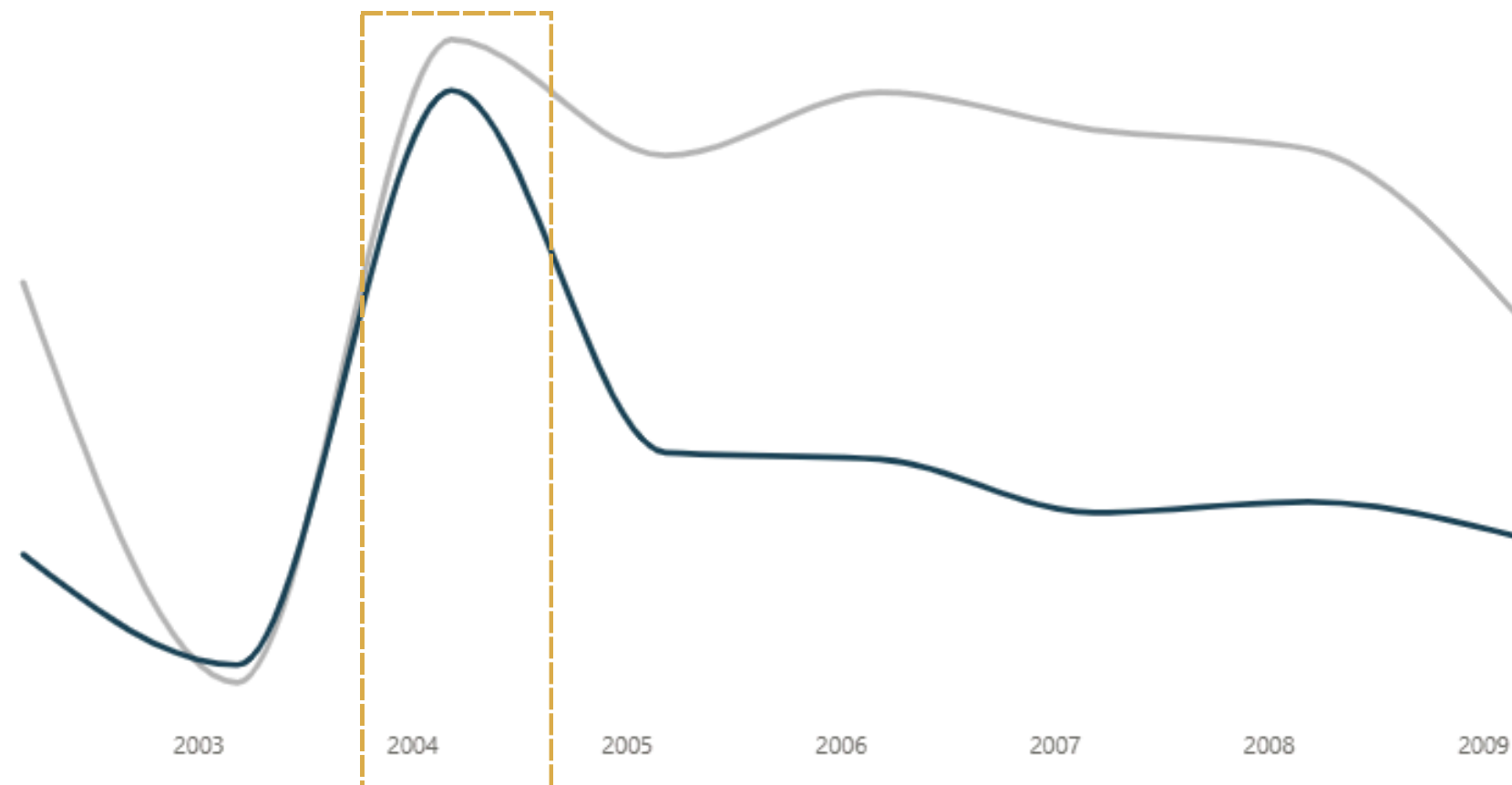


Error rate was highest for the 2004 app version and during early morning

- Error rate and the amount of time spent on each session are **directly proportional**.
- The highest error recorded was through the **2004 app version**.

Error Rate and Session Duration by App Version

● %Error Rate ● Session Duration

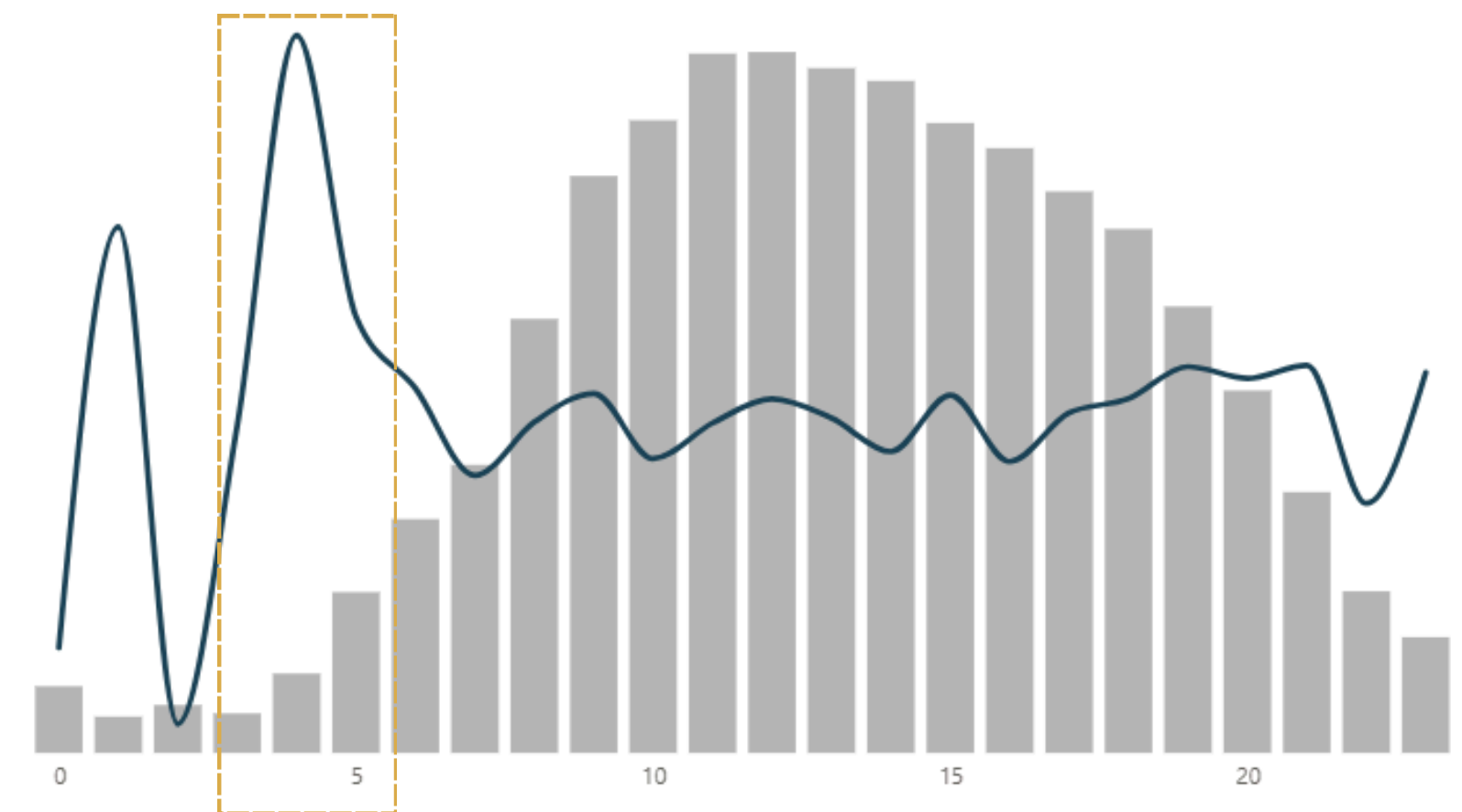


The payment error rate **reached its peak at 4 am**, although the total number of transactions was highest during noon.

→ Traffic congestion is not a problem. Rather, the problem might come from system maintenance, infrastructure, or fraud issues without timely support.

Total Transactions and Error Rate by Hour of Day

● Total Transactions ● %Error Rate

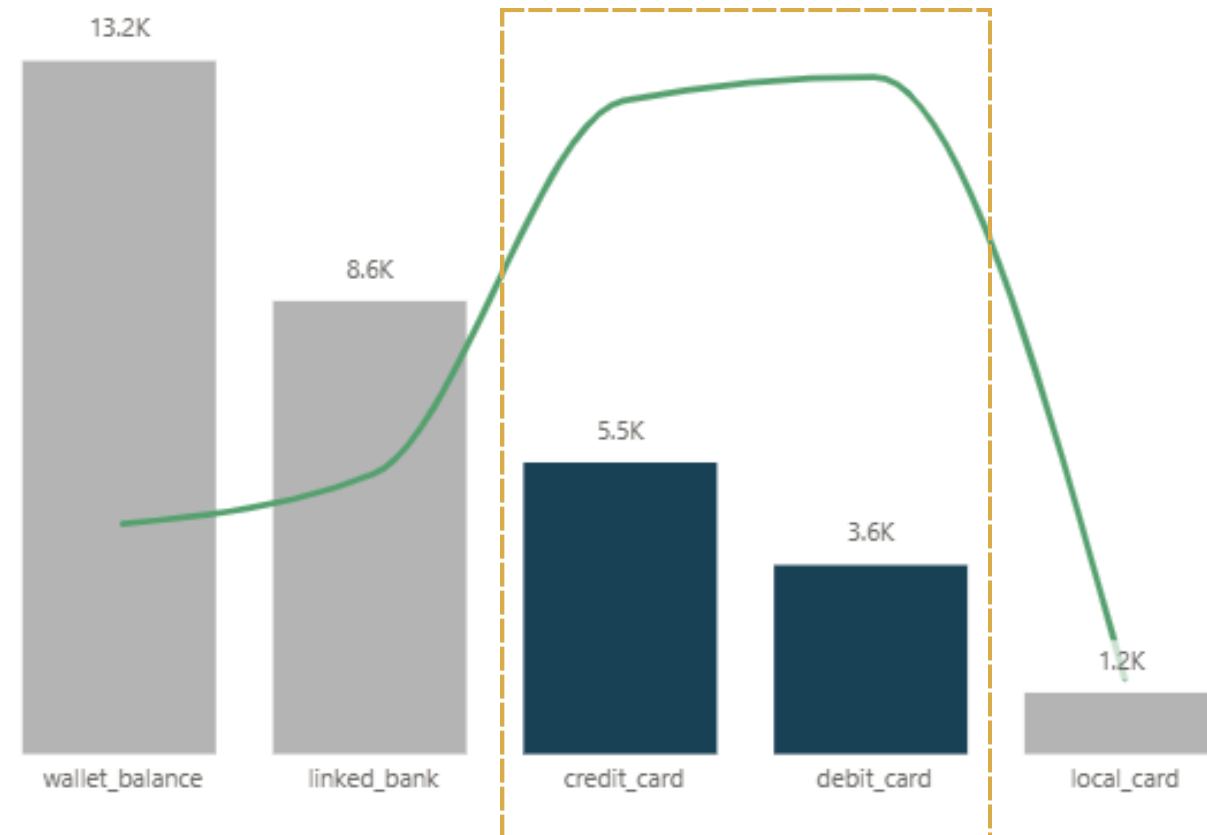


Transactions through Credit Card and Debit Card recorded the highest error rate

- Despite lower transactions than other types, **Credit Card and Debit Card** recorded the highest error rate, at **14%** respectively.
- This signals that there is a problem with payment flow through third parties (banks), consolidating that 3Ds and OTPs authentication needs to be improved.
- Making transactions by phone, through both **iOS and Android** operating systems, led to **80% errors**.
- Meanwhile, transactions through web methods, such as Payment Gateway and QR Code Scanning, are more stable with an error rate under 6%.

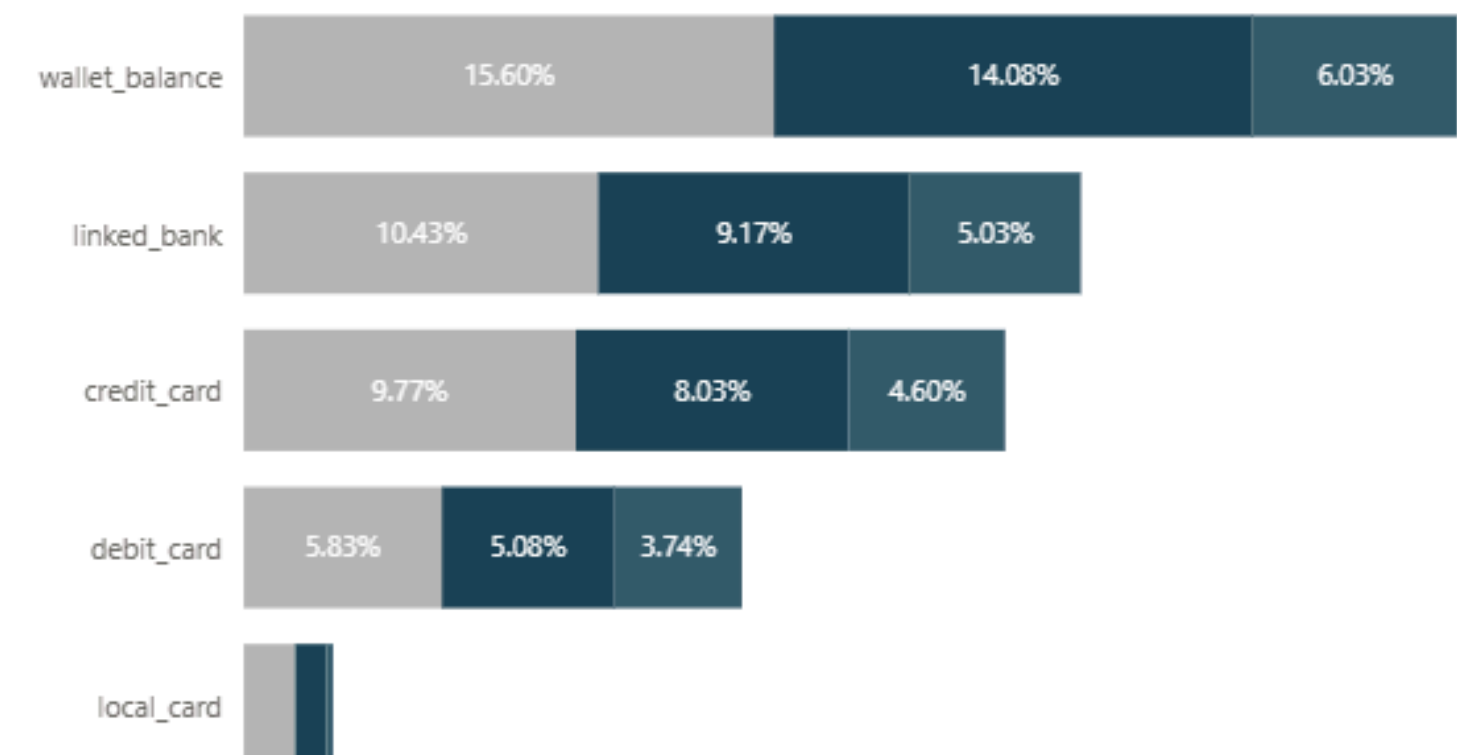
Error Rate by Payment Method

● Total Transactions ● %Error Rate



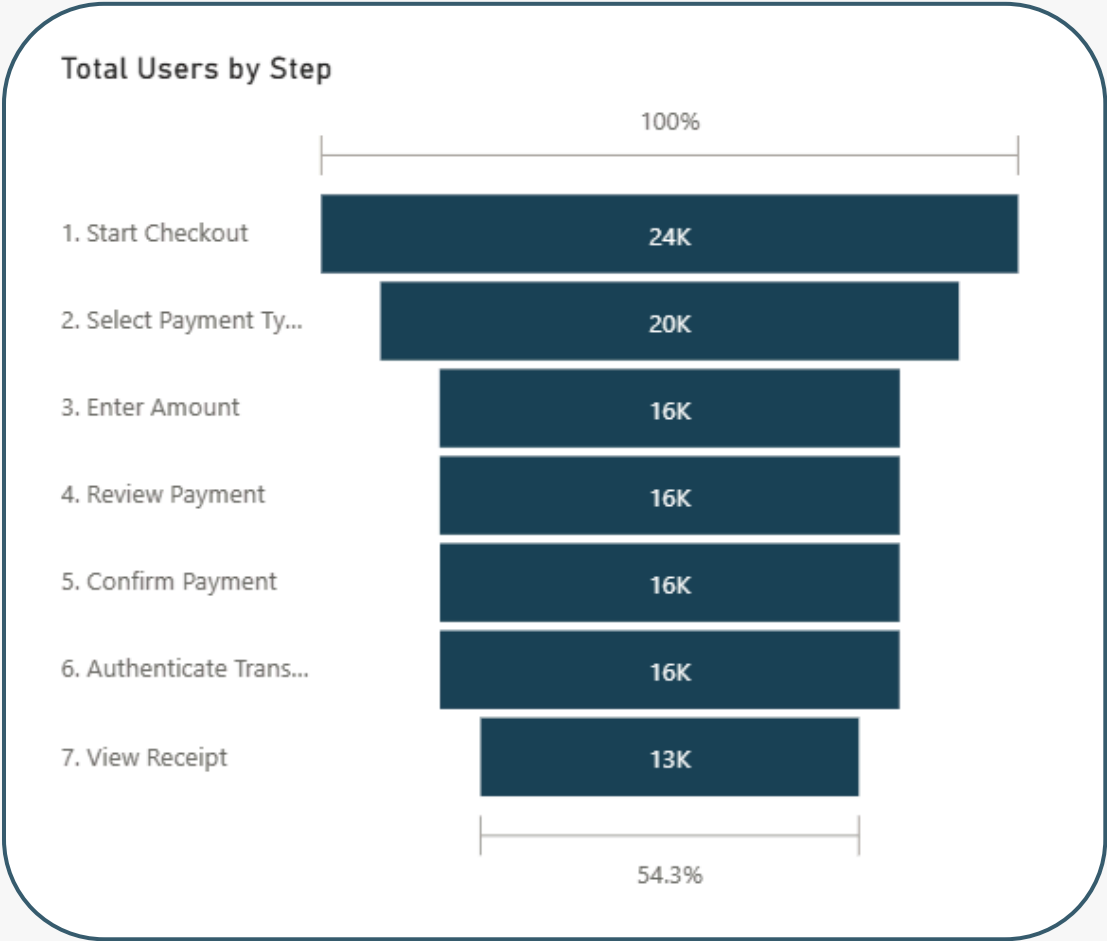
Types of error by Payment Method

platform ● android ● ios ● web



The post-authentication phase recorded the most drop-offs

- The overall conversion rate is **54.3%**, with 13K users successfully reaching the receipt stage.
- The most critical friction point occurs immediately after the app opens, where **17% of users (4K)** exit the process.
- Following the authentication phase, there was a notable **18% drop-off**. Addressing this bottleneck will boost our success rate significantly.
- The total time to convert is **141.53** seconds (~2 minutes 30 seconds).
- Users spend only **~5 seconds on Step 2**, suggesting drop-offs are likely caused by misclicks rather than UX complexity.
- The **Authentication phase has the longest average duration (31.63 seconds)**, though the difference compared to other steps is small (~2 seconds). The latest version (3.9.9) has already addressed this issue by improving payment gateway connectivity and increasing transaction timeout.



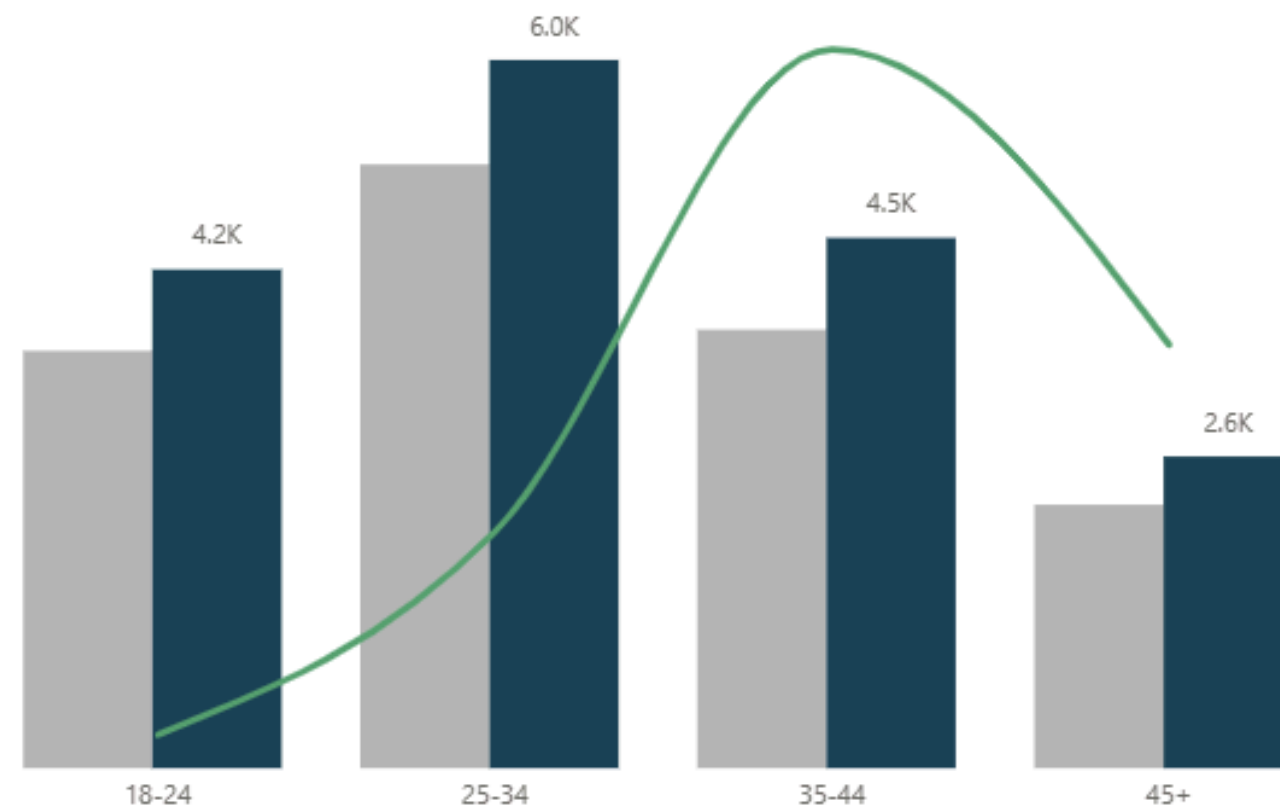
Step	Total Users	%First Step	%Previous Step	Avg time from First Step	Avg Time from Previous Step
1. Start Checkout	23,551	100%		0.00	
2. Select Payment Type	19,551	83%	83%	5.49	5.49
3. Enter Amount	15,551	66%	80%	28.80	23.28
4. Review Payment	15,541	66%	100%	42.69	13.98
5. Confirm Payment	15,532	66%	100%	71.61	28.79
6. Authenticate Transaction	15,531	66%	100%	97.12	31.63
7. View Receipt	12,784	54%	82%	141.53	27.68

The 35-44 age group and Dormant users have the highest error rate

- The **error rate peaked in the 35-44 age group** (11%), while the number of users was lower than in other age brackets.
- The **error rate** tends to **increase gradually** from the youngest group (18-24) and peaks in middle age (35-44) before decreasing in the 45+ group.
- **The most inactive (Dormant) users have the highest error rate (14%)**, indicating that transaction errors affect churn rate.
- High-activity users have the lowest error rate. They may be familiar with the transaction flow and hence, make fewer operational errors.

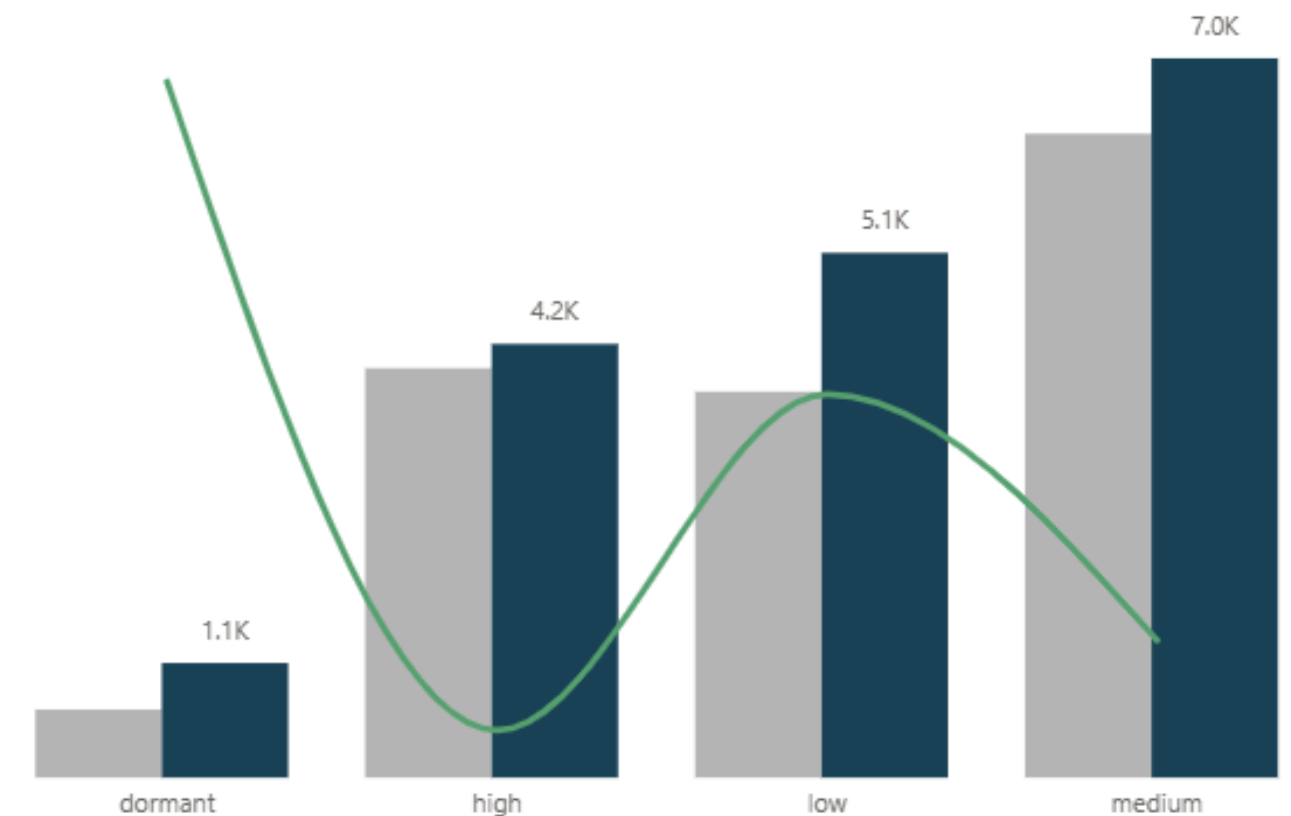
Error Rate by Age Group

● July ● August ● Error Rate (%)



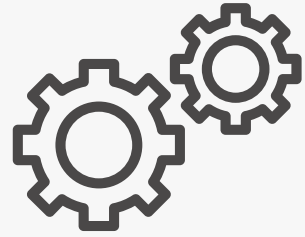
Error Rate by Activity Level

● July ● August ● Error Rate (%)



*Dormant: Users who used to use the service but stopped

Recommendations



Reduce authentication failures and backend-related drop-offs

- **Increase OTP session timeout** to reduce authentication expiration.
- Review version 3.9.9 and **fix payment gateway API bugs and backend connectivity failures** if needed.
- Review the fraud detection process to reduce false cases.
- Investigate the 2004 app version, considering suggested updates for users.
- **Conduct card-specific debugging** with billers/providers to identify root causes of Credit/Debit failures during authentication.



Payment flow optimization & Root cause validation

- **Run A/B testing** on the authentication flow (especially post-version 3.9.8 redesign).
- Compare pre- and post-redesign performance metrics to isolate impact.
- **Promote alternative payment methods** (e.g., Wallet Balance) with higher success rates while Card Payment issues are being investigated.



Real-time support and Dormant users re-engagement

- **Provide support for users aged 35+** and increase morning support, suggestively using an AI chatbot.
- In-app real-time error guidance.
- **Retarget users who dropped during authentication** (Dormant users) through email and notifications.
- Considering marketing campaigns to promote new feature updates and educate on the new payment flow.

**Thank
you for
listening**



Transactions and Success Rate by Date

● Transactions ● Success Rate

